

CASE 24

INTERFERON- γ RECEPTOR DEFICIENCY

The destruction of intracellular microorganisms in macrophages.

Certain pathogens such as mycobacteria, *Listeria*, *Leishmania*, and *Salmonella* take up residence in macrophages and are thereby protected from elimination by antibodies or cytotoxic T cells. These microorganisms can be eliminated only when their host macrophages are activated and produce increased amounts of nitric oxide, oxygen radicals, and other microbicidal molecules (Fig. 24.1). The activation of macrophages in an adaptive immune response is masterminded by T_H1 cells (Fig. 24.2); the most important cytokine involved in macrophage activation is interferon (IFN)- γ .

When macrophages take up microorganisms by phagocytosis, they secrete interleukin (IL)-12. This cytokine is necessary for the induction of IFN- γ synthesis by T cells and natural killer (NK) cells. Furthermore, IL-12 favors the maturation of T_H1 cells, which activate macrophages, and suppresses the maturation of T_H2 cells, which secrete a cytokine, IL-10, which suppresses activation of inflammatory macrophages.

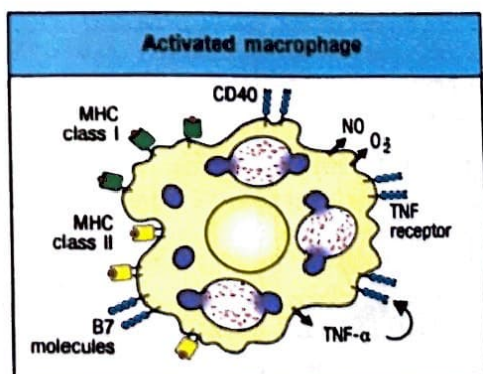


Fig. 24.1 Activated macrophages undergo changes that greatly increase their antimicrobial effectiveness and amplify the immune response. Once the macrophage is activated, lysosomes fuse with the intracellular vesicles within which the pathogenic bacteria (red) reside, which exposes the microorganisms to degradative enzymes and other microbicidal agents. Activated macrophages also increase their expression of receptors for tumor necrosis factor (TNF), and secrete TNF- α . This autocrine stimulus synergizes with interferon (IFN)- γ secreted by T_H1 cells to increase the antimicrobial action of the macrophage, in particular by inducing the production of nitric oxide (NO) and oxygen radicals ($O_2^{\cdot -}$). The macrophage also increases the expression of CD40, which by interaction with the CD40 ligand on T cells upregulates the expression of B7 proteins and increases the expression of class II MHC molecules on the macrophage, thus allowing the further activation of resting CD4 T cells.

TOPICS BEARING
ON THIS CASE:

Microbicidal action
of phagocytes

Macrophage
activation

Cytokine receptor
signaling pathway

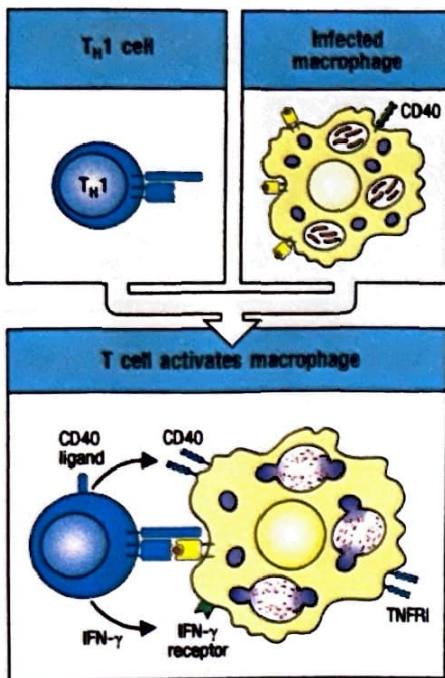


Fig. 24.2 T_H1 cells activate macrophages to become highly microbicidal. When a T_H1 cell specific for a bacterial peptide contacts an infected macrophage, the T cell is induced by IL-12 to secrete the macrophage-activating factor IFN- γ and to express CD40 ligand. Together, these newly synthesized T_H1 proteins activate the macrophage.

2½-year-old female child with enlarged lymph nodes. Order blood test.

Fig. 24.3 Diagram of the IFN- γ receptor and its activation of the JAK-STAT signaling pathway. A dimer of IFN- γ causes clustering of the four chains of the IFN- γ receptor. The receptor 1 chains become phosphorylated by JAK1, forming a site to which the transcription factor STAT1 α can bind. This in turn becomes phosphorylated, dimerizes, and is then transported to the nucleus, where it upregulates various genes, leading to macrophage activation. In the figure, the red dots identify phosphorylated residues. The IFN- γ R1 chain is encoded on chromosome 6, whereas the IFN- γ R2 chain is encoded on chromosome 21.

IFN- γ binds to a receptor expressed on the membrane of macrophages. This receptor is composed of two different polypeptide chains—IFN- γ receptor 1 (IFN- γ R1) and IFN- γ R2, each of which is associated with a tyrosine kinase—JAK1 and JAK2, respectively. When a dimer of IFN- γ binds to two molecules of IFN- γ R1, it causes them to associate with two IFN- γ R2 chains (Fig. 24.3) and this initiates a signaling pathway that eventually results in changes in gene transcription.

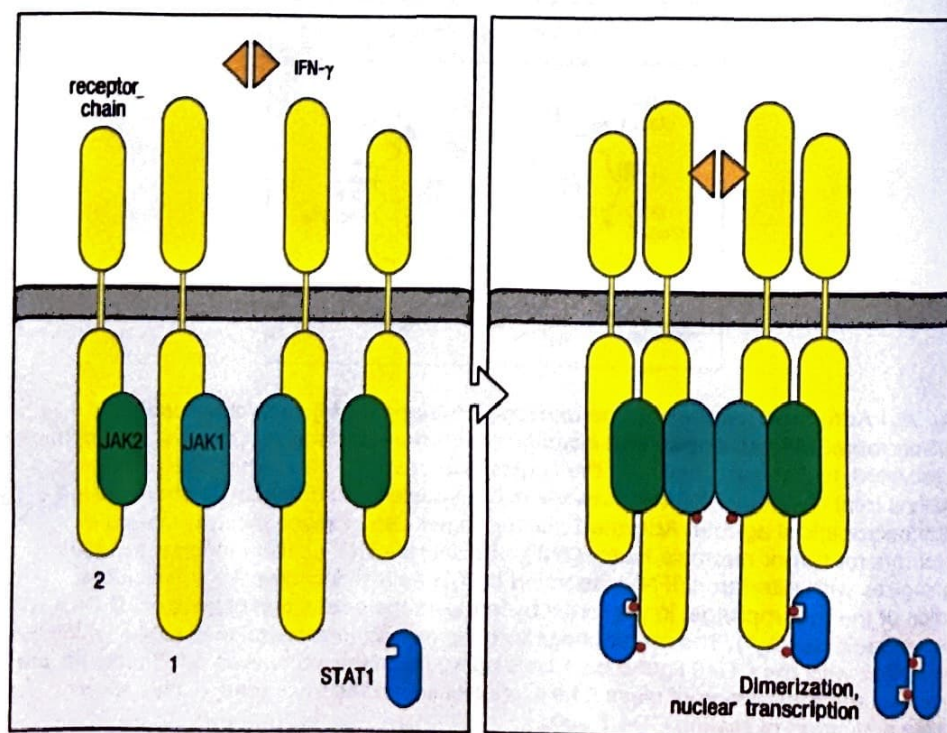
As we shall see in this case, a mutation in genes that encode for proteins along the IL-12/IFN- γ axis have drastic effects on the capacity to fight certain pathogens, such as environmental mycobacteria.

The case of Clarissa Dalloway: a relentless infection due to mycobacteria.

Clarissa Dalloway was the first child born to a couple who lived in an isolated fishing village on the coast of Maine. Her parents believed that they were distantly related to each other. The fishermen of this village were all descended from English settlers who came there in the late 17th century, and there was much intermarriage in the community.

Clarissa was well at birth and developed normally until she was around 2½ years old. Her mother then noticed that she was not eating well, had diarrhea, and was losing weight. She took Clarissa to a pediatrician in the town of Bath, Maine. The pediatrician, Dr Woolf, noted enlarged lymph nodes and ordered an ultrasound and CT scan of the chest and abdomen. These showed enlarged lymph nodes in the mesentery and para-aortic region, and Dr Woolf referred Clarissa to the Children's Hospital in Boston.

Blood tests revealed a white blood cell count of $9400 \mu\text{L}^{-1}$, of which 55% were neutrophils, 30% lymphocytes, and 15% monocytes (slightly elevated). Her serum IgG was 1750 mg dL^{-1} , IgA 450 mg dL^{-1} , and IgM 175 mg dL^{-1} (these immunoglobulin values are all elevated).



A decision was made to biopsy the enlarged lymph nodes. Histological examination showed marked increase in the number of histiocytes and macrophages, and many neutrophils. There was no granuloma formation and no giant cells were seen. An acid-fast stain for mycobacteria (see, for example, Fig. 46.4) revealed numerous microorganisms, and *Mycobacterium avium intracellulare* was cultured from the lymph nodes as well as from the blood.

Despite appropriate antibiotic treatment for the mycobacterial infection, Clarissa eventually developed infiltrates in the lungs and progressive enlargement of the spleen. At 6 years old she developed sepsis, and *Salmonella paratyphi* was cultured from her blood. She was successfully treated with antibiotics for this infection, but soon afterward she developed meningitis and died. *M. avium intracellulare* was cultured from the cerebrospinal fluid.

A detailed family history revealed that Clarissa had three male cousins, two of whom were brothers, who had died of mycobacterial infections. In one case *M. chelonae* had been cultured from lymph nodes; *M. fortuitum* and *M. avium intracellulare* had been cultured from blood and lymph nodes of the two brothers.

Do lymph node biopsy.

Expanded number of macrophages; mycobacterium infection.

Interferon- γ receptor deficiency.

It is estimated by the World Health Organization that approximately 2 billion human beings currently alive have been infected with mycobacteria. Yet only a very small fraction of these individuals develop disease as a result of this infection. The AIDS epidemic has markedly increased the incidence of mycobacterial disease due to *M. tuberculosis*, and to other mycobacterial species such as *M. avium intracellulare*, which are collectively called 'atypical mycobacteria' or 'environmental mycobacteria.' In general, atypical mycobacteria do not cause disease in immunologically-competent human beings, except for swollen lymph nodes in which these mycobacteria survive. Infection with atypical mycobacteria causes a positive tuberculin reaction to develop.

Although there are many ethnic differences in susceptibility to *M. tuberculosis*, no single genetic factor had ever been found in humans to explain this susceptibility until Clarissa Dalloway and her family were found to have a genetic defect in the IFN- γ R1 gene (*IFNGR1*). The genetic defect was ascertained in this family by mapping the susceptibility gene to chromosome 6q22, the map location of the receptor gene.

This finding prompted the examination of children who had developed progressive mycobacterial infection after immunization with the BCG (*Bacillus Calmette-Guérin*) vaccine, and who did not have severe combined immunodeficiency (see Case 5). More kindreds with the IFN- γ R1 deficiency were discovered as a result of this lead. Subsequently, a similar susceptibility to mycobacterial infection has been found in patients with defects in genes encoding the IFN- γ R2 chain, the p40 subunit shared by IL-12 and IL-23, or the IL-12 receptor (IL-12R) β 1 chain. Thus the dependence of IFN- γ synthesis on IL-12, and the critical role of the IL-12/IFN- γ axis in controlling mycobacterial disease have been neatly confirmed by these human mutations. Furthermore, causes of Mendelian susceptibility to mycobacterial disease (MSMD) also include mutations of STAT1 (a transcription factor that is phosphorylated upon IFN- γ -dependent activation of JAK1/JAK2 (see Fig. 24.3), and that mediates the expression of IFN- γ -responsive genes), IRF8 (a transcription factor induced by IFN- γ), ISG15 (a potent inducer of IFN- γ synthesis), and NEMO (IKK- γ), which regulates the activation of NF κ B (see Case 23) and hence controls the production of IL-12.

IFN- γ receptor deficiency can be inherited either as an autosomal recessive or autosomal dominant trait, and may be complete or partial, depending on the nature of

the genetic defect. IL-12 and IL-12R β 1 deficiency are autosomal recessive diseases. Dominant-negative, heterozygous *STAT1* mutations cause MSMD; in contrast, complete *STAT1* deficiency is inherited as an autosomal recessive trait and causes MSMD associated with severe susceptibility to viral infections. IRF8 deficiency may be inherited as an autosomal recessive or as an autosomal dominant trait; the former leads to increased susceptibility to a broad range of pathogens, whereas autosomal dominant IRF8 deficiency is responsible for MSMD. ISG15 deficiency is an autosomal recessive disease. Finally, NEMO deficiency is an X-linked disorder.

It is interesting to note that disseminated mycobacterial infections may also be due to neutralizing autoantibodies directed against IFN- γ . This condition is more common in patients from Southeast Asia, and emphasizes the important role played by IFN- γ in anti-mycobacterial immunity.

Questions.

- 1 How do you explain the alarming increase in the incidence of mycobacterial infections in patients with AIDS?
- 2 Clarissa Dalloway and her cousins had positive tuberculin skin tests. Would you have predicted this?
- 3 Clarissa and her cousins also did not develop granulomas as a result of infection with atypical mycobacteria. On one occasion mycobacteria were surprisingly cultured from Clarissa's blood. What do these observations tell us?
- 4 Clarissa also had a problem with salmonella, but she had no problem with pneumococcal infection or with any viruses, such as chickenpox. How do you explain this?

CASE 25

SEVERE CONGENITAL NEUTROPENIA

A lack of neutrophils, leading to difficulty in fighting infections.

Neutrophils are phagocytic cells that play a crucial role in the response to many types of infections, especially infections involving extracellular bacteria. These innate immune cells are the first cells to be recruited to sites of infection and inflammation, into which they migrate in large numbers from the bloodstream. Once activated, neutrophils are short-lived; after ingesting microbes and killing them, neutrophils die and are broken down by the action of macrophages, producing the pus that is characteristic of many extracellular bacterial infections (Fig. 25.1). Neutrophils recognize their microbial targets mostly through invariant receptors, such as Fc γ receptors and Toll-like receptors. The binding of antibody-coated ('opsonized') microorganisms to Fc γ receptors on the neutrophil cell surface

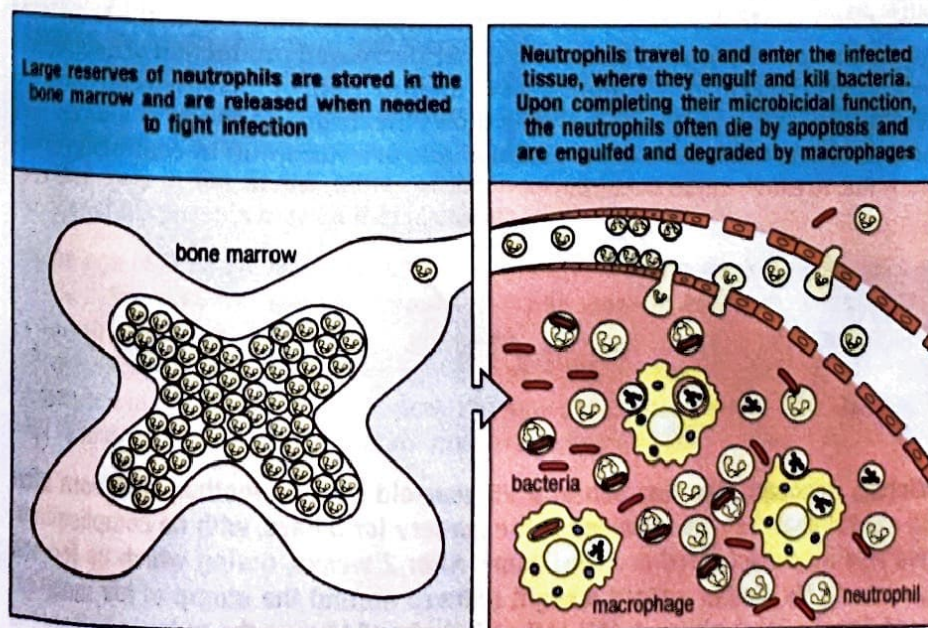


Fig. 25.1 Neutrophils are stored in the bone marrow and move in large numbers to sites of infection, where they act and then die. After one round of ingestion and killing of bacteria, a neutrophil dies. The dead neutrophils are eventually mopped up by long-lived tissue macrophages, which break them down. The creamy material known as pus is composed of dead neutrophils.

TOPIC BEARING ON
THIS CASE:

Neutrophil function